

CHERAW, SC, USA

WMO: 720605

Lat: 34.717N

Lon: 79.950W

Elev: 240

StdP: 14.57

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 00197

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.3	24.6	-0.2	5.5	32.4	4.9	7.2	33.4	28.6	53.7	21.7	54.1	2.8	260	0.384

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.4	95.0	74.5	92.9	74.3	90.8	74.1	78.6	88.9	77.8	87.4	76.9	85.8	6.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.4	134.3	81.6	75.1	132.7	81.1	73.5	125.7	79.2	42.2	89.2	41.1	88.0	40.4	84.6	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.3	15.2	12.4	14.5	98.4	4.3	2.2	11.4	100.0	8.9	101.3	6.5	102.6	3.4	104.2		
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.0	42.0	46.7	51.8	62.2	70.7	77.7	80.1	78.6	73.6	62.6	51.2	46.1
	DBStd	15.38	9.97	10.53	9.02	7.60	6.36	4.42	3.48	3.91	5.40	7.62	7.77	9.29
	HDD50	827	288	174	89	6	0	0	0	0	0	4	75	191
	HDD65	2967	715	517	418	141	27	0	0	0	6	135	418	590
	CDD50	5217	39	83	144	372	642	830	934	888	709	394	111	71
	CDD65	1882	1	6	8	57	204	380	469	423	266	60	4	4
	CDH74	17416	4	41	108	597	1802	3718	4534	3866	2158	556	24	8
	CDH80	7413	0	1	13	137	649	1722	2147	1741	861	141	1	0
Wind	WSAvg	4.8	4.9	5.6	5.7	5.7	4.7	4.4	4.3	3.8	4.4	5.0	4.5	4.2
Precipitation	PrecAvg	46.0	4.0	3.6	4.1	2.9	3.2	4.6	5.3	5.1	3.9	3.3	2.8	3.0
	PrecMax	62.5	8.0	8.3	9.0	7.0	6.9	10.1	10.3	10.8	13.1	12.2	6.6	7.4
	PrecMin	29.7	0.8	0.7	0.6	0.0	0.8	0.6	0.6	0.4	0.3	0.0	0.7	0.2
	PrecStd	6.8	1.8	1.9	1.9	1.7	1.6	2.5	2.3	2.6	2.6	2.6	1.5	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.2	79.1	81.6	86.5	94.8	98.8	98.6	95.3	93.3	89.6	77.3	74.8
		MCWB	61.8	63.7	60.8	64.9	72.5	74.7	75.3	75.0	73.1	76.0	63.9	67.4
	2%	DB	67.6	73.4	76.9	83.8	90.1	94.5	95.0	93.0	90.7	83.5	72.8	71.6
		MCWB	58.6	59.9	59.8	65.3	70.7	73.8	75.2	74.8	72.3	69.6	62.1	65.7
	5%	DB	63.5	68.5	72.9	81.2	86.3	91.2	92.7	91.0	88.4	80.8	69.9	66.2
		MCWB	56.4	57.8	57.2	64.5	70.0	73.4	75.2	75.2	72.0	67.2	59.9	63.4
	10%	DB	58.9	64.2	68.3	77.1	83.6	89.9	90.5	89.8	85.6	76.6	65.8	63.1
		MCWB	52.6	55.9	55.8	62.4	69.5	73.0	74.9	75.0	71.5	66.0	58.8	59.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.9	67.0	66.1	71.8	75.7	78.3	80.7	79.9	77.8	76.7	68.5	70.3
		MCDB	69.4	73.5	72.2	78.3	85.2	90.4	92.4	89.2	87.3	83.3	71.9	72.4
	2%	WB	62.3	64.4	63.4	69.9	74.0	77.1	78.9	78.6	76.3	74.3	66.0	67.0
		MCDB	65.2	69.8	68.2	76.3	82.7	88.6	89.9	88.1	84.7	79.6	69.9	69.4
	5%	WB	58.4	61.7	61.6	67.9	72.7	76.0	77.9	77.8	75.0	71.8	63.5	63.8
		MCDB	61.6	68.5	68.9	75.4	81.3	86.5	87.7	86.7	82.3	76.9	65.7	64.6
	10%	WB	54.0	57.9	59.2	65.8	71.4	75.0	76.9	76.7	73.5	68.4	60.6	59.3
		MCDB	57.5	63.7	66.8	73.3	80.4	84.6	85.7	84.9	79.5	73.7	65.5	61.4
Mean Daily Temperature Range	5% DB	MDBR	20.8	21.9	23.1	24.3	21.6	21.6	20.4	19.7	20.6	23.0	23.8	19.6
		MCDBR	25.9	26.3	28.9	27.7	25.7	24.3	22.9	22.1	24.6	26.5	26.6	21.8
		MCWBR	17.4	15.4	14.5	11.9	9.3	7.9	7.4	7.2	8.9	12.5	16.7	15.7
	5% WB	MCDBR	20.8	23.3	23.2	21.1	20.1	21.6	19.7	18.7	20.0	18.4	19.4	19.3
		MCWBR	17.1	16.1	14.6	11.5	8.5	8.1	7.4	7.0	8.2	11.1	14.9	16.4
Clear-Sky Solar Irradiance	taub	0.316	0.326	0.362	0.405	0.467	0.491	0.543	0.515	0.432	0.389	0.337	0.312	
	taud	2.503	2.469	2.399	2.268	2.146	2.119	1.998	2.065	2.287	2.374	2.477	2.547	
	Ebn at Noon	283	292	289	279	263	255	241	246	262	265	272	278	
	Edh at Noon	26	30	36	43	49	50	56	51	39	33	27	24	
All-Sky Solar Radiation	RadAvg	855	1060	1405	1760	1931	2032	1923	1747	1492	1228	938	744	
	RadStd	48	117	106	127	156	141	119	82	127	151	76	65	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (73 neighbors)	+0.94	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+306	+223	

Nomenclature: See separate page